

Module – IV**12 Hrs**

AWT: AWT Classes window fundamentals, component, container, panel, Window, Frame , Canvas, Creating a frame window in an Applet , working with Graphics ,Control Fundamentals , Layout managers, Handling Events by Extending AWT components. Core java API package, reflection, Remote method Invocation (RMI)

Swing: J applet, Icons & Labels, Text fields, Buttons, Combo boxes, Tabbed panes, Scroll panes, Trees, Tables.

Event Handling and JDBC: Delegation Event model, Event Listener Interfaces, jdbc

JavaFX: Graphics, User Interface Components, Effects, Animation, and Media, Application Logic, Interoperability, JavaFX Scene Builder 2

Text Books:

1. Programming in Java. Second Edition. OXFORD HIGHER EDUCATION. (SACHIN MALHOTRA/SAURAV CHOUDHARY)
2. CORE JAVA for Beginners. (Rashmi Kanta Das), Vikas Publication

Reference Books:

1. JAVA the Complete Reference (9th Edition) Herbalt Schelidt.
2. Programming with Java: Bhave &. Patekar, Pearson Education.
3. Big Java: Horstman, Willey India

PCCS4204	Formal Language and Automata Theory	3-0-0	Credit-3
Module 1 (10 Hours) Alphabet, languages and grammars. Production rules and derivation of languages. Chomsky's hierarchy of languages and Grammars. Regular grammars, regular expressions and finite automata (deterministic and nondeterministic). Closure and decision properties of regular sets. Pumping lemma of regular sets. Minimization of finite automata. Left and right linear grammars. DFA/NFA to regular expression and vice versa using Arden's Formula. Module 2 (10 Hours) Context free grammars and pushdown automata. Chomsky and Griebach normal forms. Parse trees, Cook, Younger, Kasami, and Early's parsing algorithms. Ambiguity and properties of context free languages. Pumping lemma, Ogden's lemma, Parikh's theorem. Deterministic pushdown automata, closure properties of deterministic context free languages.			

PCCS4203	Object Oriented Programming Using JAVA	3-0-0	Credit-3
Course Objectives 1. Gain knowledge about basic Java language syntax and semantics to write Java programs and use concepts such as variables, conditional and iterative execution methods etc. 2. Understand the fundamentals of object-oriented programming in Java, including defining classes, objects, invoking methods etc ,exception handling mechanisms, inheritance, packages and interfaces			

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B.Tech Syllabus for Admission Batch 2018-19 Onwards

Computer Science & Engineering

3. Understand the principles of multithreading, Applet etc.
4. Develop java based software application using awt, swing, event driven interfaces and jdbc.

Course Outcomes

1. knowledge of the structure and model of the Java programming language, (knowledge)
2. use the Java programming language for various programming technologies (understanding)
3. develop software in the Java programming language, (application)
4. evaluate user requirements for software functionality required to decide whether the Java programming language can meet user requirements (analysis)
5. propose the use of certain technologies by implementing them in the Java programming language to solve the given problem (synthesis)
6. choose an engineering approach to solving problems, starting from the acquired knowledge of programming and knowledge of operating systems. (evaluation)

Module – I

10Hrs

Introduction: -Object oriented programming, Features, Pillars and Advantages of OOP ,java, Features of java, advantages of java over C and C++,JDK, JRE, JVM, Byte code

Fundamentals of Programming: - Data Types, variable, Modifiers, Typecasting,

Operators and their precedence, Decision making and looping, one dimension arrays and its application in searching and sorting process, multi dimension arrays and matrix operations, Receive input using Scanner class, Buffer Reader, Command line arguments..

Objects and classes: -class, object, methods, parameter passing, Constructor, object passing, constructor overloading, nested classes, static, final, this keyword

Module - II

08 Hrs

Inheritance: - simple and multilevel inheritance, Using Super to Call Super class constructor, Method overriding, Dynamic method Dispatch, Using Abstract Classes, Using final with inheritance.

Packages & Interfaces: Packages, Access Protection, Importing package, Interface, Implementing Interfaces, variables in Interfaces, Interfaces can be extended.

Exception Handling: Exception and its types, Error, Exception handling, Using try& catch, Multiple catch, throw, throws, finally, Java's Built in exceptions, user defined exception.

Module - III

10 Hrs

Multithreading: Java Thread Model, Thread Priorities, Creating a thread, Creating Multiple threads, Using is Alive () and join (), wait () & notify (), Deadlock, Synchronization

String Handling: String constructors, String length, Character Extraction, String Comparison, Modifying a string.

Applets: Basics, Architecture, Skeleton, The HTML APPLET Tag, Passing parameters to Applets, Applet context and show documents ().

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Registration no:

2001105213

Total Number of Pages: 02

B.Tech.

Subject Code: PCCS4203

59-60

4th Semester Regular/Back Paper Examination 2021-22
 Subject Name: Object Oriented Programming Using JAVA
 Branch: Computer Science & Engg.

Time: 3 Hours

Max Marks: 60

Question Code:

Answer Question No.1 which is compulsory and any five from the rest.
 The figures in the right-hand margin indicate marks.

Q8

Write short notes on any two:

- ✓ a) Constructors
- ✓ b) Abstract class
- c) Multipath Inheritance
- d) Interfaces

5 × 2 = 10

- Q1 Answer the following questions: (1x10)
- ✓ a) Differentiate between Object and Object Reference.
 - ✓ b) What is an inheritance? Name the inheritance not supported by Java.
 - ✓ c) What is a package? And explain its importance.
 - ✓ d) How to declare arrays in Java?
 - ✓ e) What is the use of "this" keyword?
 - ✓ f) What is a thread?
 - ✓ g) _____ class objects are mutable.
 - ✓ h) What is an Autoboxing?
 - ✓ i) Objects can be synchronized by using _____ method and _____ keyword.
 - ✓ j) What is a method overriding?
- ✓ Q2 a) Using class and object, print the largest and smallest of n-integers. (5)
- ✓ b) What are the different possible ways for creation of String Objects? Explain with examples. (5)
- ✓ Q3 a) Using class and object, check a string is palindrome or not. (5)
- ✓ b) Explain dynamic method dispatch with a suitable example. (5)
- ✓ Q4 a) Write a command-line argument program that will display the sum and average of integers. (5)
- ✓ b) Explain super and super() with suitable examples. (5)
- ✓ Q5 a) Explain constructor overloading with a suitable example. (5)
- ✓ b) Write a command-line argument program that will display the largest and smallest of any three integer numbers. (5)
- ✓ Q6 a) Explain thread state diagrammatically. (5)
- ✓ b) Write a Java program that will execute multiple tasks using a single thread. (5)
- ✓ Q7 a) Write a Java program that will add two times using the object as a return value kind. The format of the time is hours:minutes:seconds. (5)
- ✓ b) Explain the methods used for blocking a thread. (5)

Registration no:

2 1 0 1 1 0 5 5 0 7



Total Number of Pages: 02

B.Tech.

Subject Code: PCCS4203

4th Semester Regular/ Back Paper Examination 2022-23
Subject Name: Object Oriented Programming using JAVA
Branch: Computer Science & Engineering

Time: 3 Hours

Max Marks: 60

Question Code:

3 / 10



Answer Question No.1 which is compulsory and any five from the rest.
The figures in the right-hand margin indicate marks.

Q1

Answer the following questions:

(1x10)

- String and String Buffer both in Java is a _____?
i) class ii) Object iii) Literals iv) variables
- _____ method is used to invoke an overridden method?
- _____ method of String class is used to obtain the index value of 2nd or 3rd occurrence of a character in string with the help of _____?
- _____ modifier provides maximum security and _____ lowest?
- _____ type of conversion reflects the condition in which size of destination datatype is less than source datatype?
- _____ represents the attributes and _____ represents the behavior of an object?
- _____ class is extended to create custom Exception?
- Method used to capture and handle ItemEvent is _____?
- _____ method is used to draw a circle using applet?
- _____ life cycle states of applet through which it will move only once during the life cycle of applet?

Q2

- What do you mean by command line arguments? Why and in which case it will be used?

(2+3)

Write a program to find factorial of difference between average of all even and odd numbers in a collection of 10 numbers using command line arguments? Do not use array?

- What do you mean by type conversion? Mention various approaches used for this?

(2+3)

Write a java program to difference between Sum of product of consecutive even digits except 2 and 6 and Sum of product of consecutive odd digits except 3 and 7 of any digit number?

Q3

- Method overloading is possible by simply changing its return type. True or False? Justify your answer?

(2+3)

Write a program to override sum method if arguments are numbers than it will add numbers or string than concat two strings using method overloading?



(1+4)

- b) Class is a template of an object. True or False? Justify your answer?
Write a program to perform following using class and object for deposit, withdraw, check balance for banking application?
I. Assign a fixed minimum balance Rs. 3000 for account.
II. Deposit user defined amount and display updated balance. If balance is less than minimum balance then bank will charge 10% of amount user want to deposit as penalty and rest will be deposited to the account.
III. Withdraw user defined money if user has sufficient fund otherwise show insufficient fund.
IV. Check for balance.
V. Maintain a fixed amount Rs. 3000 in the account.

- Q4 a) Mention the names of life cycle states through which a thread will go for multiple times during its lifetime instead of one and why? (1+4)
Write a program to perform following using multi threading and priority without synchronization?
a) find greatest among 3 numbers
b) find sum of product of corresponding digits of two any digit number. Such as $n=1234$ $m=7896$ output= $6*4+9*3+8*2+7*1$

- b) What are the modifiers that cannot be used with interface and why? (1+4)
Write a java program to compute following series using simple inheritance and interface?
 $1+x^2/2! + x^4/4! + x^6/6! + \dots + x^n/n!$
Where x = Sum of product of consecutive digits of num
 n = Sum of product of consecutive prime digits of num

- Q5 a) Exception and Error both are same or different? Justify your answer? (1+4)
Write a java program to create invalid rate of interest exception for bank if Rate of Interest (ROI) provided by bank like SBI, UBI and HDFC are not in the range of Rate of Interest (ROI) provided by RBI? If ROI is valid then find simple interest of the desired bank using simple inheritance and abstract class?

- b) What are the differences between abstract class and interface? (3+2)
Write a program to find division of two numbers using try, catch and finally block?

- Q6 a) Which component of AWT is used to create applet? Write a java program to create a scrolling message from right to left using applet and multi threading? Message will be taken from user about their hobbies using JOptionPane? (1+4)
b) Define Applet? Describe its life cycle with neat and clean diagram? (3+2)
Write a java program to draw a circle inside a square using Applet?

- Q7 a) What is AWT? How it differs from swing? Which one is better and why? (3+2)
What are the components of delegation event models? Explain its working principle by using the components?
b) Write a java program to create simple calculator to find summation, subtraction, division, multiplication and modulus of two numbers using JLabel, JTextField, and ActionListener? (5)

- Q8 Write short notes on any two: (5x2)
a) Dynamic Method Dispatch
b) Modifiers
c) Package
d) Constructor overloading

Registration No.													
		INDIRA GANDHI INSTITUTE OF TECHNOLOGY, SARANG 4th Semester Regular/ Back Paper Examination AY 2023-24								Total no. of Pages: 02			
Branch	CSE						Programme	B.Tech.					
Subject Name	Object Oriented Programming using JAVA						Semester	4th					
Subject Code	PCCS4203						Year	2023-24					
Time: 03 Hrs	Answer Question number 1 which is compulsory and any five from the rest.						Maximum Marks	60					
Each Question Carry Marks(M) as Indicated in the right-hand Margin													
Q. No.	Questions	M	B.L	C.O	PO								
1. a	Classname always starts with _____ letter and method starts with _____ letters?	01	L2										
b	The default datatype for floating point number is _____ and its size is _____.	01	L1										
c	_____ concept is used to implement runtime polymorphism?	01	L1										
d	_____ concept is used to invoke an overridden method in java?	01	L1										
e	_____ access specifier cannot be used with interface?	01	L1										
f	_____ class cannot be inherited?	01	L2										
g	Throws keyword is used with _____ exception?	01	L3										
h	_____ method makes sure that all the threads will get a fair chance to execute and the thread will terminate?	01	L2										
i	_____ method of _____ listener is used to handle textevent?	01	L1										
j	_____ lifecycle state of the applet occurred multiple times during its lifespan?	01	L1										
2. a	What is conditional operator? Write a java program to find difference between Sum of product of consecutive even digits except 2 and 6 and Sum of product of consecutive odd digits except 3 and 7 of any 4 digit number using conditional operator and without using looping and decision making?	01 + 04	L3										
b	Which type of data is supported by command line arguments? Write a java program to compute following series and take input num using command line arguments. $1 - x^2/2! + x^3/3! - x^4/4! + \dots + x^n/n!$ Where x= Sum of product of consecutive digits of num n= Sum of product of consecutive even digits of num	01 + 04	L3										
3. a	What is the mechanism used to assign a key to each and every item of array? Write a program to calculate following of student having 4 subjects C, C++, Java, Python. Mark of each subject should be inbetween 0 to 100. i) take mark of each subject using scanner class	01 + 04	L3										

	ii) Find total mark and %age of marks obtained ii) Find grade of each subject				
b	Constructor returns nothing. True or False. Justify your answer? Write a program to find volume of three boxes in which dimension of first box is known to user, the second box is double of first box and the third box is the sum of first and second box using constructor overloading with class and object?	01 + 04	L3		
4 . a	Method can be overloaded by changing the return type only. True or False. Justify your answer? Write a program to check whether a number or string is palindrome or not using method overloading?	01 + 04	L3		
b	Abstract class is better than interface. True or False. Justify your answer? Write a program to find area, perimeter and volume of a Rectangular, square and Circular Box using multilevel inheritance, abstract class and method overriding.	01 + 04	L4		
5 . .	Execution of Finally block can be skipped. True or False. Justify your answer? Write a program to perform following using interface, package with custom exception to deposit, withdraw, check balance for banking application i) Assign a fixed value Rs. 2500 as minimum balance for account ii) Deposit user defined amount if balance is more than minimum balance otherwise InsufficientFunddepositException occurred and alert the user that 10% of the deposited money will be deducted as pealty and rest in credited to your account and finally display updated balance. iii) Withdraw user defined money if user has sufficient fund otherwise show InsufficientfundException using custom exception to alert the user that sufficient fund not available for withdraw. Also check after winthdraw operation if the resultant balace is less than minimum balance than 5% of remining balance will be deducted as penalty for each debit opeartion. iv) Check for balance v) Maintain a fixed amount Rs. 2500 in the account	01 + 09	L5		
6 . a	Synchronization is an unavoidable part of multithreading. True or False. Justify your answer? Write a program to program to print your first name and last name using synchronization?	01 + 04	L4		
b	Which method is used to call the paint method of applet? Write a program to print your a message received from user using JoptiojnPane that is blinking and scrolling from left to right using Applet and multithreading?	01 + 04	L5		
7 . a	Swing is preferred over AWT. True or False. Justify your answer? Write a program to find summation, subtraction, multiplication, division, modulus and power of 2 numbers using swing and event handling?	01 + 04	L2		
b	What is event handling? Explain the delegation event model with neat and clean architecture? Write a program to create one application for colour recognition for kids in which kids will chose button that shows colour name like RED,BLUE, etc. and after clicking on button background colour of window will change as per colour name?	2. 5 + 2. 5	L6		
Short notes (Answer any Two)					
8 . a	Exception vs Error	05	L1		
b	Interface vs Abstract class	05	L1		
c	Throw vs Throws	05	L1		
d	AWT hierarchy	05	L1		

		INDIRA GANDHI INSTITUTE OF TECHNOLOGY, SARANG 4 th Semester Regular/ Back Paper Examination AY 2024-25		Total no. of Pages: 02	
Branch	Computer Science & Engineering	Program	B. Tech		
Subject Name	Object Oriented Programming using JAVA	Semester	4 th		
Subject Code	PCCS4203	Year	2024-25		
Time: 03 Hrs	Answer Question number 1 which is compulsory and any five from the rest.	Maximum Marks	60		
Each Question Carry Marks (M) as Indicated in the right-hand Margin					
Q's	Questions	M	BL	CO	PO
1.a	_____ modifier make sure that variable is resource is accessible within the package only?	01	L1	1	1
b	_____ method provides maximum number of character a string buffer can hold?	01	L1	3	3
c	Constructor returns nothing. True or False?	01	L1	2	2
d	_____ can be used only as a base class?	01	L1	2	2
e	_____ approach is used to call an overridden method?	01	L1	2	2
f	_____ approach provides 100% abstraction?	01	L1	3	3
g	_____ method make sure that thread will terminate?	01	L1	3	3
h	_____ method is used to call paint() method of applet?	01	L1	4	4
i	_____ component is used to create Applet window?	01	L1	4	4
j	_____ is used to control the appearance and layout of Component over container?	01	L1	4	4
2.a	Write a program to compute following series using the number num taken from user through command line arguments? $1-x^2/2! + x^4/4! -x^6/6! + \dots + x^n/n!$ Where x= Sum of product of consecutive digits of num n= Sum of product of consecutive prime digits of num	05	L3	1	1
b	What is command line arguments? Write a java program to find difference between Sum of product of consecutive even digits except 2 and 6 and Sum of product of consecutive odd digits except 3 and 7 of any 4 digit number using conditional operator ? Take the number from user using command line arguments. Do not use decision making or looping.	05	L3	1	1
3.a	What is the difference between Checked Exception and Unchecked Exception? Write a program to find division of two numbers using try catch and finally block?	05	L4	3	3
b	Method overloading is used to implement compile time polymorphism and Why? Write a program to find sum of two numbers, sum of two strings and sum of two complex numbers using method overloading?	05	L4	2	2
4.a	Which keyword is used to access the method of super class? Write a program to find volume, weight and cost of a box using multilevel inheritance and interface of 2 boxes in which dimension of 1 st box is entered by user whereas 2 nd box in 5times of 1 st box? Weight is k times of volume in Kg. and cost is Rs. 5/Kg.	05	L5	2	2

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b	Which Modifiers can be applied with interface and Why? Write a program to perform following using interface and package to deposit, withdraw, and check balance for banking application I. Assign a fixed value as balance for account II. Deposit user defined amount and display updated balance III. Withdraw user defined money if user has sufficient fund otherwise show Insufficient fund using custom exception IV. Check for balance V. Maintain a fixed amount 3000 in the account	05	L5	3	3
5.a	What are the two basic purpose of using package? Write a program to create InvalidDevisionOperationException that will occur whenever the denominator is prime and palindrome and factorial of numerator and also find division of two numbers in which denominator should not be prime and palindrome and factorial of numerator using InvalidDevisionOperationException using interface and package?	05	L6	3	3
b	Final class cannot be extended. True or False? Justify your answer? Write a program to find area and perimeter of Rectangle and Square using method overrrding and abstract class?	05	L4	2	2
6.a	Define Multithreading? Mention its significance in java programming? Write a program to create 3 threads in which first thread will print firstname, second thread will print middle name and third thread will print last name using Synchronization, isAlive and Join with Priority 9 for First thread, 8 for Second Thread, and 7 for 3rd Thread?	05	L5	3	3
b	What is Deadlock? Explain the process od deadlock detection and recovery using suitable examples?	05	L2	3	3
7.a	What are the lifecycle methods of applet that invoked only once during its lifecycle? Write a program to move a ball in L shape using applet and multithreading?	05	L3	4	4
b	AWT is preferred over Swing. True or False. Justify your answer? Write a program to find greatest among three numbers using JLabel, JtextField and JButton with event handling?	05	L3	4	4
ANSWER ANY TWO in Question 8					
Short notes					
8.a	Delegation Event Model	05	L 1	4	4
b	Modifier	05	L1	1	1